

Astragaloside IV protects against ischemic brain injury in a murine model of transient focal ischemia

Yumin Luo^{a,*}, Zhen Qin^a, Zhen Hong^a, Xinmin Zhang^b, Ding Ding^a, Jian-Hui Fu^a,
Wei-Dong Zhang^c, Jun Chen^d

^aInstitute of Neurology, Fudan University Huashan Hospital, Shanghai 200040, PR China

^bInstitute of Integrated Traditional Chinese (TC) and Western Medicine, Fudan University Huashan Hospital, Shanghai 200040, PR China

^cCollege of Pharmacy, Second Military Medical University, Shanghai 200433, PR China

^dState Key Laboratory of Medical Neurobiology, Fudan University, Shanghai 200040, PR China

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Abstract

Astragalus membranaceus is a herbal medicine that has been used clinically in stroke patients in China for decades, but its potential neuroprotective effect against ischemic brain injury has not been experimentally tested. In this study, we investigated the effect of *Astragaloside IV*, a purified extract from *Astragalus membranaceus*, in a murine model of focal cerebral ischemia/reperfusion produced by transient (1.5 h) middle cerebral artery occlusion. As determined at 72 h after ischemia, post-ischemic treatment of *Astragaloside IV* (20 or 40 mg/kg) markedly and significantly ($P < 0.03$ vs. vehicle-treated animals) reduced infarct volume. *Astragaloside IV* treatment also decreased the levels of malondialdehyde, an indicator of lipid peroxidation, and increased the levels of the antioxidant enzymes glutathione peroxidase and superoxide dismutase in ischemic tissues. The results presented here provide the first evidence of a neuroprotective effect of *Astragaloside IV* in the model of ischemic brain injury. We suggest that the anti-infarction effect by *Astragaloside IV* may be derived at least in part from its antioxidant properties.

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It is well known that post-ischemic reperfusion enhances the formation of reactive oxygen species (ROS) in brain tissues [4,9,11]. Excessive production of ROS, such as superoxide anion, hydroxyl radical, hydrogen peroxide, and nitric oxide (NO), is believed to play a critical role in the development of ischemic/reperfusion brain injury [2]. ROS may contribute to brain injury by directly attacking macromolecules, including proteins, lipids and DNA, or indirectly through affecting cellular signaling pathways and gene regulation [3, 15,19]. ROS can be scavenged partially by endogenous antioxidants, including superoxide dismutase (SOD) that catalyzes the dismutation of the superoxide anion, glutathione peroxidase (GSH-PX) and catalase (CAT) that mediate the breakdown of hydrogen peroxide [5,6]. Therefore, an enhancement of antioxidant activities in brain

tissues may be potentially beneficial for neuronal recovery from ischemic/reperfusion injury.

With the successful application of thrombolytic therapy such as tissue plasminogen activator in clinical stroke, there are strong needs for the development of neuroprotectants with antioxidant properties that are particularly suitable for the treatment of the ischemia/reperfusion type of brain injury. A number of Chinese medical herbs have been shown to possess strong antioxidant properties [7]. One such herb, *Astragalus membranaceus*, has been routinely used in China for patients with stroke or chronic debilitating diseases, and even for normal subjects who wish to further improve their vital functions. A pilot clinical investigation suggested that *Astragalus membranaceus* is safe and may be beneficial for the treatment of acute cerebral infarction [1]. However, at the present time there is no direct experimental evidence confirming that *Astragalus membranaceus* or its active constituents offer neuroprotection against ischemic brain injury.

* Corresponding author. Tel.: +86-21-6248-9999, ext. 6216; fax: +86-21-6248-3421.

E-mail address: yumin_l@hotmail.com (Y. Luo).

The present study was undertaken to investigate the potential protective effect of *Astragaloside IV*, the major active constituent of *Astragalus membranaceus*, in a murine model of focal cerebral ischemia/reperfusion. *Astragaloside IV* (3-O-beta-D-xylopyranosyl-6-O-beta-D-glucopyranosyl-cycloastragenol, molecular structure shown in Fig. 1) is a purified small molecular weight (MW 784) saponin. In vitro studies have demonstrated that *Astragaloside IV* is a strong scavenger for superoxide radicals ($IC_{50} \approx 0.023$ mg/ml) and hydroxyl radicals ($IC_{50} \approx 0.053$ mg/ml) [10]. The antioxidant properties of *Astragaloside IV* are likely derived from the phenyl hydroxyl group and cyclic propane group in the saponin molecule [10,20]. The *Astragaloside IV* used in this study was of high purity (99.2%) as determined by HPLC analysis, and the compound was provided by College of Pharmacy, Second Military Medical University, Shanghai, China.

All experimental procedures were performed under protocols approved by the Animal Care Committee of the Animal Center at the Chinese Academy of Sciences in Shanghai and in accordance with the principles outlined in the NIH *Guide for the Care and Use of Laboratory Animals*. A total of 77 adult male C57/B6 mice, weighting 25–30 g, were obtained from the Animal Center at the Chinese Academy of Sciences in Shanghai, and housed in a light- and temperature-controlled room. The experiments consisted of two parts: the first part was to evaluate the effect of *Astragaloside IV* on infarction induced by focal ischemia and reperfusion; the second part was to study the effect of *Astragaloside IV* on ischemia-induced changes in cerebral tissue contents of malondialdehyde (MDA), an indicator of lipid peroxidation, and the activity levels of the antioxidant enzymes GSH-PX and SOD.

For the first part of the study, 45 mice were randomly assigned into one of the three experimental groups ($n = 15$ per group): the vehicle group in which saline was injected, and drug groups 1 and 2 which received *Astragaloside IV* (dissolved in saline) at doses of 20 and 40 mg/kg, respectively. In the drug treatment groups, each animal

received three *Astragaloside IV* injections (i.p.): immediately after middle cerebral artery occlusion (MCAO), and then 24 and 48 h after MCAO. In the vehicle group, saline was injected using the same time schedule. For the second part of the study, 33 mice were randomly divided into four experimental groups ($n = 8$ per group), including the three ischemic groups as described above, and the sham-operation group, in which ischemia was not induced.

Transient focal cerebral ischemia (1.5 h duration) was induced using a previously described procedure [22]. In brief, the mice were anesthetized by intraperitoneal injection of chloral hydrate (dissolved in distilled water) at a dose of 300 mg/kg. Rectal temperature was recorded and maintained at 37 ± 0.5 °C throughout the surgical procedure and up to 2 h after reperfusion. A midline neck incision was made and, under an operative microscope, the left common carotid artery (CCA), external carotid artery (ECA), and internal carotid artery (ICA) were exposed. A 5.0 monofilament nylon thread (Ethicon, Johnson and Johnson), with its tip rounded by heating briefly near a flame, was inserted into the ECA and then advanced into the lumen of the ICA to occlude the origin of the middle cerebral artery (MCA). The nylon filament remained in the lumen for 1.5 h, and was then withdrawn to allow the initiation of reperfusion. The local cerebral blood flow in the cortex ipsilateral to MCAO was monitored during ischemia and the first 30 min of reperfusion using the laser-Doppler flowmeter (Perimed Perflux4001, Sweden). The animals were returned to cages after they awoke from anesthesia. The animals that could not normally intake food and water in the second day after surgery or that developed seizures after surgery were terminated and excluded from further studies.

All mice were sacrificed 72 h after MCAO or after sham operation. For the first part of the study, the animals were killed in a CO₂ chamber, and the brains were quickly removed and sectioned into six consecutive coronal slices of 2 mm thickness. The slices were stained by immersing into 2% of 5-triphenyl-2H-tetrazolium chloride (TTC) for 30 min at 37 °C, and then fixed with 8% of formalin. The infarct volume of each brain was measured using the UTHSCS ImageTool 200 software.

For the measurements of MDA, SOD and GSH-PX, brain tissues were quickly removed from skull and kept in a -80 °C freezer until subjected to analysis. The content of MDA was measured as described by Ohkawa et al. [14]. Briefly, brain tissues were homogenized in 10 times (w/v) 0.1 M sodium phosphate buffer (pH 7.4). To 0.1 ml of the above sample, the following reagents were added: acetic acid (20%) 1.5 ml (pH 3.5), thiobarbituric acid (0.8%) 1.5 ml, and sodium dodecyl sulfate (8.1%) 0.2 ml. The mixture was heated at 100 °C for 60 min, and then cooled with tap water. After the addition of 5 ml of *n*-butanol/pyridine (15:1 v/v) and 1 ml distilled water, the reaction mixture was shaken vigorously. After centrifugation at 4000 rev./min for 10 min, the organic layer was withdrawn and absorbance was

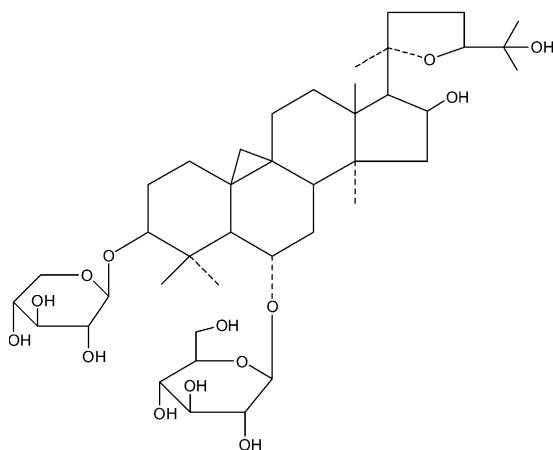


Fig. 1. Structure of *Astragaloside IV*. The molecular weight is 784.

measured at 532 nm using a spectrophotometer (Spectronic Instruments Inc., NY, USA).

The activities of antioxidant enzymes were measured using commercially available kits (Nanjing Bioengineering, Nanjing, Jiangsu Province, PR China). The assay for SOD activity was based on its ability to inhibit the oxidation of oxymine by O_2^- produced from the xanthine-xanthine oxidase system. One unit of SOD activity was defined as the capability of reducing the absorbance at 550 nm by 50%. The assay of GSH-PX activity was determined by quantifying the rate of oxidation of the reduced glutathione (GSH) to the oxidized glutathione (GSSG) by H_2O_2 catalyzed by GSH-PX. One unit of GSH-PX was defined as the activity that reduces the level of GSH by $1 \mu\text{mol/l}$ in 1 min per mg protein.

All data were reported as mean $\pm$ SD. The Kruskal–Wallis variance analysis was used for comparing independent groups. $P < 0.05$ was found, therefore for intergroup comparisons, the Mann–Whitney U -test with Bonferroni correction was applied, and $P < 0.05$ was accepted as significant in the first part of the study since there were three groups (Fig. 2). On the other hand, $P < 0.03$ was accepted as significant for the remaining comparisons since there were four independent groups and the results are represented in Figs. 3–5.

The first objective of the present study was to determine the effect of *Astragaloside IV* treatment on cerebral infarction induced by focal ischemia and reperfusion. Infarct volume was measured at 72 h after MCAO, a time point at which infarction reaches maturation in this model [13,16]. In a pilot study ($n = 7$ per group), we found no significant protection by *Astragaloside IV* at 5 mg/kg (data not shown). Therefore, in this study we tested *Astragaloside IV* at 20 and 40 mg/kg, respectively. As shown in Fig. 2, *Astragaloside IV* at these doses offered significant protection against the formation of cerebral infarction ($P < 0.03$ vs. the vehicle-treated group), reducing infarct volume by 30% and 32% at 20 and 40 mg/kg, respectively. In all animals, the core body temperature was controlled in the normal range, thus excluding the potential hypothermic effect from the tested compound. Moreover, as determined using a laser-Doppler flowmeter, at either dose tested *Astragaloside IV* treatment did not alter cortical blood flow after ischemia (data not shown), suggesting that the protection by this compound was independent of improved cerebral blood perfusion during MCAO.

The second objective of this study was to determine whether the anti-infarction protective effect by *Astragaloside IV* is associated with any antioxidant effect as suggested by others based on in vitro observations. Accordingly, we measured the tissue contents of MDA, an indicator for lipid peroxidation, and the activity levels of the antioxidants SOD and GSH-PX, comparing between vehicle- and *Astragaloside IV*-treated animals. As shown in Fig. 3, the contents of MDA were significantly increased in the vehicle-treated group at 72 h after MCAO ($P < 0.01$ vs. sham non-ischemic

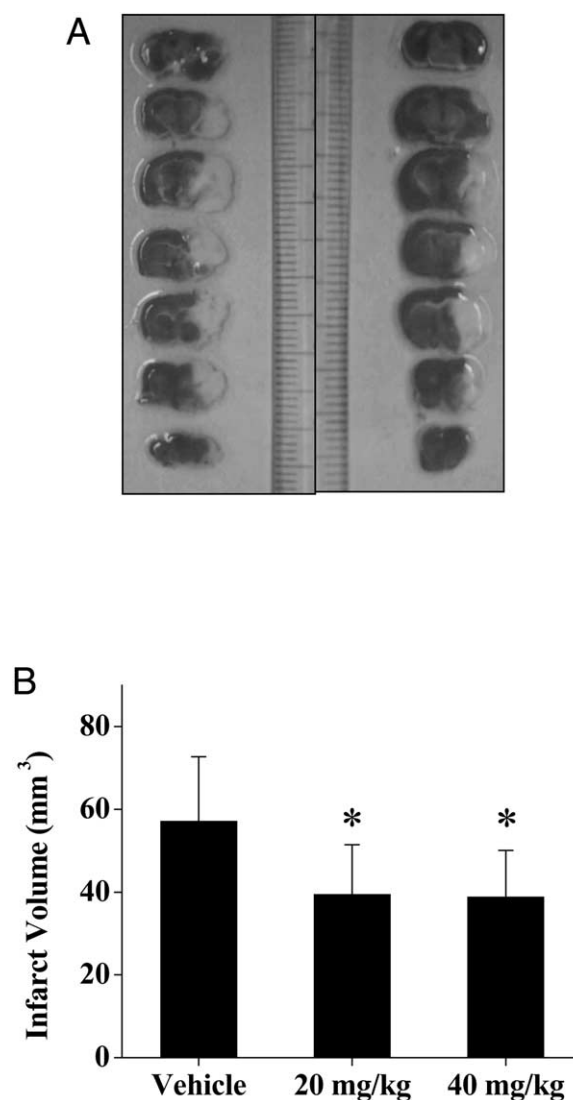


Fig. 2. The effects of *Astragaloside IV* on the cerebral infarction. (A) The two pictures are TTC staining for the mice brain following ischemia/reperfusion. The left picture is of the vehicle-treated group. The right picture is of 40 mg/kg administered *Astragaloside IV*. (B) Effect of *Astragaloside IV* (20 and 40 mg/kg) on the infarction volume after 72 h reperfusion in the mice MCAO model. Data represent the mean $\pm$ SD ($n = 15$ per group). The infarct volume was markedly reduced in the two drug-treated groups (* $P < 0.03$ vs. the vehicle-treated group).

controls), whereas *Astragaloside IV* treatment at either dose completely attenuated the increases in MDA contents after ischemia. Consistent with this antioxidant effect, *Astragaloside IV* treatment also significantly restored cerebral tissue levels of SOD (Fig. 4) and GSH-PX (Fig. 5) activity. In vehicle-treated animals, the levels of SOD and GSH-PX activities were decreased by 27% and 30% ($P < 0.01$ vs. sham non-ischemic controls) after ischemia/reperfusion. However, in *Astragaloside IV*-treated groups, the levels of SOD and GSH-PX activity were recovered to near non-ischemic control levels, and these changes reached statistic significance as compared to the vehicle-treated groups (Figs. 4 and 5). These effects by *Astragaloside IV* treatment

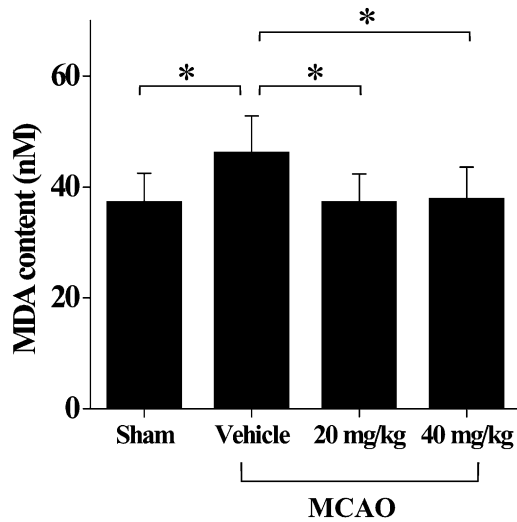


Fig. 3. Effects of *Astragaloside IV* on MDA content of the brain tissues at 72 h after MCAO in mice. Data represent the mean $\pm$ SD ($n = 8$ per group). The content of MDA was significantly increased in the vehicle-treated group at 72 h after MCAO ($*P < 0.01$ vs. sham non-ischemic controls). The contents of MDA were markedly decreased in the *Astragaloside IV* treatment group at either dose ($*P < 0.01$ vs. the vehicle-treated group, respectively).

on the levels of antioxidants were unlikely via a direct effect on the activity or expression of the enzymes, as *Astragaloside IV* had no effect on SOD or GSH-PX activity in sham non-ischemic controls or in the non-ischemic hemisphere contralateral to MCAO (data not shown).

Despite a long history in China of using *Astragalus*

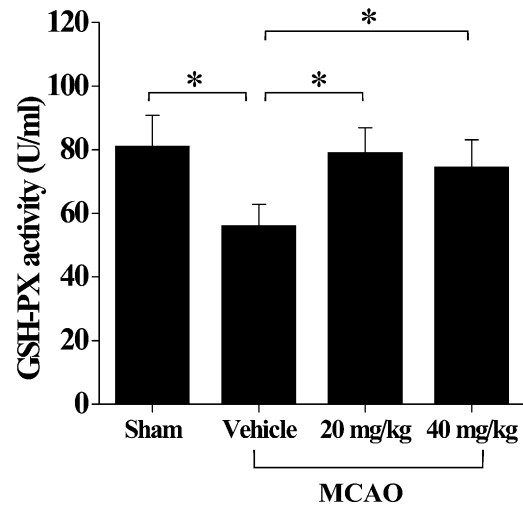


Fig. 5. The effects of *Astragaloside IV* on the activity of GSH-PX of ischemic brain tissue at 72 h after MCAO. Data represent the mean $\pm$ SD ($n = 8$ per group). The activities of GSH-PX were significantly decreased in the vehicle-treated group ($*P < 0.01$ vs. sham non-ischemic controls). The activities of GSH-PX were increased in the *Astragaloside IV*-treated groups ($*P < 0.01$ vs. the vehicle-treated group, respectively).

membranaceus as a therapeutic agent in clinical stroke and the experience-based perception that such treatment might be beneficial, there is no clear experimental evidence supporting this speculation. The data presented in this report provide the first pharmacological evidence demonstrating that *Astragaloside IV*, a purified active constituent from *Astragalus membranaceus*, offers significant neuroprotec-

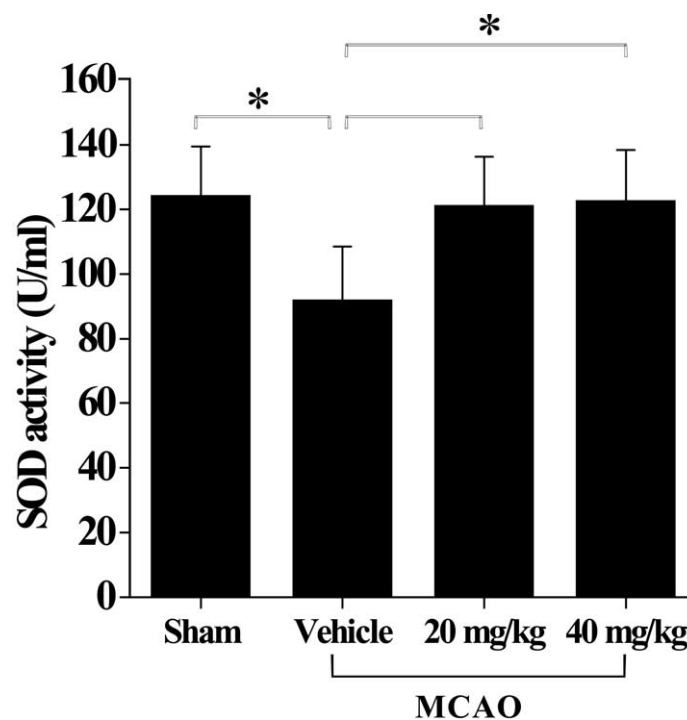


Fig. 4. The effects of *Astragaloside IV* on the level of SOD of ischemic brain tissue at 72 h after MCAO in mice. Data represent the mean $\pm$ SD ($n = 8$ per group). The levels of SOD were significantly decreased in the vehicle-treated group ($*P < 0.01$ vs. sham non-ischemic controls). The levels of SOD were increased in the *Astragaloside IV*-treated groups ($*P < 0.01$ vs. the vehicle-treated group, respectively).

tion against the formation of cerebral infarction in a murine model of focal ischemia/reperfusion. The protective effect by *Astragaloside IV* was achieved without altering the core temperature of the animals or the cortical blood flow after MCAO, the two major physiological factors that could substantially affect the outcome of ischemic brain injury.

An antioxidant effect may be one of the underlying mechanisms by which *Astragaloside IV* protects against ischemic/reperfusion brain injury. *Astragaloside IV* is a potent free radical scavenger capable of reducing both superoxide and hydroxyl radicals [10]. Although the precise biochemical process of the interaction between *Astragaloside IV* and ROS is not understood, it has been suggested that the phenyl hydroxyl and cyclic propane groups are likely responsible for the antioxidant properties of *Astragaloside IV* [10,20]. Consistent with its role as an antioxidant, we observed that treatment of *Astragaloside IV* attenuated the increases in the cerebral contents of the lipid peroxidation marker MDA after ischemia/reperfusion, and restored the tissue levels of SOD and GSH-PX. This effect may be secondary, due to the decreased accumulation of ROS in tissues by *Astragaloside IV*, and, consequently, decreased consumption of antioxidant enzymes. The observation that *Astragaloside IV* had no effect on SOD or GSH-PX activity in non-ischemic brain tissues does not support the alternative possibility that *Astragaloside IV* may directly enhance the activities of antioxidant enzymes in the brain.

Another pharmacological property of *Astragaloside IV* that could contribute to its neuroprotective effect in ischemic brain injury is the anti-inflammatory activity. Several in vivo and in vitro studies have demonstrated the anti-inflammatory effect of *Astragaloside IV*. For instance, in cultured human umbilical vein endothelial cells, *Astragaloside IV* potently inhibits leukocyte adhesion to endothelial cells by decreasing cytokine- or LPS-induced expression of adhesion molecules such as E-selectin and VCAM-1 [24]. In murine peritoneal macrophages, *Astragaloside IV* inhibits the production and release of the inflammatory mediators interleukin-1 and TNF- α [21]. These potential anti-inflammatory effects of *Astragaloside IV* in CNS injury have not been investigated thus far, but their investigation is clearly warranted. There is substantial evidence that inflammation, such as neutrophil infiltration and release of interleukin-1 and TNF- α , contributions to secondary brain injury after ischemia and reperfusion [8], and pharmacological anti-inflammatory treatments have shown effectiveness in focal ischemia/reperfusion models [17,18,23]. One of the downstream cytotoxic effects of neutrophil recruitment to the area of ischemia/reperfusion is the overproduction of ROS [12]. Accordingly, it is possible that *Astragaloside IV* may achieve its neuroprotective effect in the present study by antagonizing the two main cytotoxic events of ischemia/reperfusion injury: oxidative stress and inflammation. Nevertheless, future studies are needed to further address this hypothesis.

In sum, the data presented in this report demonstrate that the active constituent of the Chinese herb *Astragalus membranaceus*, *Astragaloside IV*, is neuroprotective in the murine model of focal cerebral ischemia/reperfusion after systemic administration. The protective effect of *Astragaloside IV* is, at least in part, derived from its antioxidant properties. We suggest that purified *Astragaloside IV* may be a promising therapeutic agent for the treatment of clinical stroke.

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